

# Md Abu Sayed

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## SUMMARY

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Ph.D. candidate bridging deep learning, robotic vision, simulation, and medical imaging. Developer of NavySim (multi vessel autonomy & naval simulator) and ThreatMap (maritime threat visualization), with peer-reviewed publications in both autonomous systems and healthcare AI. Committed to cross-disciplinary collaboration and translational research.

## EDUCATION

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### University of Nevada, Reno

Reno, NV

*Ph.D., Computer Science and Engineering*

*Aug 2021 – Present*

- **Major Courses:** Introduction to Machine Learning, Fundamentals of Deep Learning, Stochastic Deep Learning, Mass Detection in Mammograms, Advanced Bioinformatics, Autonomous Mobile Manipulation, etc.
- **Dissertation:** Deep Generative and Explainable Learning for Intent Recognition in Multi-Agent Systems

### University of Nevada, Reno

Reno, NV

*M.S., Computer Science and Engineering*

*Aug 2021 – Dec 2023*

- **Thesis:** Threatmap: A Framework for Enhancing Security Awareness and Decision-Making for Naval Agents

### Khulna University

Khulna, Bangladesh

*B.S., Computer Science and Engineering*

*Dec 2014 – Jan 2019*

- **Thesis:** Automated method to segment retinal blood vessels from color fundus photographs.

## RESEARCH INTERESTS

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Intent Recognition in Multi-agent Systems; Temporal Deep Learning, Generative Modeling, and Simulation & AI; Embodied Human-robot Collaboration; Medical Image Analysis, and Computer Vision.

## TEACHING INTERESTS

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**Plan, Activity, and Intent Recognition**, Machine Learning (introductory and advanced), Deep Learning, Algorithms, Data Structures, Artificial Intelligence, Python and C/C++, and Simulation/Game Development.

## EXPERIENCE

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### Graduate Research Assistant

Aug 2021 – Present

*University of Nevada, Reno*

*Reno, NV*

- **Office of Naval Research (ONR)** Project: Graduate Researcher on ONR-funded project for maritime autonomy (2022–25), developing early intent recognition and threat-aware decision support in naval contexts.
- **On-water Deployment Collaborator:** Huntington Ingalls Industries (HII) on autonomous naval systems.
- Developing ML models for real-time prediction of vessel behavior under sparse, noisy, and adversarial conditions.
- Collaborating on simulation-driven experimentation, dataset design, and integration of ML/DL models into maritime situational awareness frameworks.
- Contributed to a pending **collaborative external grant proposal** (ONR and ONR Global) with my PI and Flinders University, drafting technical sections on temporal intent modeling and simulation-supported evaluation.

### Graduate Teaching Assistant

Aug 2021 – Present

*University of Nevada, Reno*

*Reno, NV*

- Assisted instruction in **CS 477/677: Analysis of Algorithms** (4 semesters) and **CS 422/622: Introduction to Machine Learning** (1 semester), supporting student learning in algorithmic reasoning and applied ML.
- Led review sessions and held weekly office hours to provide conceptual guidance, problem-solving support, and individualized clarification of core course topics for students across diverse backgrounds.
- Designed and graded homework, exams, and programming tasks, delivering constructive and actionable feedback.

### Lecturer

Jun 2019 – Jul 2021

*The Millennium University*

*Dhaka, Bangladesh*

- **Undergraduate Courses:** Structured Programming, Data Structure, and Artificial Intelligence & Neural Networks, etc.
- Designed lecture materials, assignments, quizzes, and examinations aligned with learning outcomes
- Supervised project-based learning activities emphasizing software design patterns, testing, and documentation.
- Served as Course Coordinator, contributing to course scheduling and alignment of curriculum across instructors.

- Developing **ICONIQ**, an AI-driven experience recommendation platform; and **AIJINTSEE**, a marketing automation system using FastAPI, PostgreSQL/pgvector, ML orchestration, and LLM & DL models integration.
- Leading and mentoring a team of 4 students (2 M.S., 2 B.S.) on architecture design, coding standards, and experiment design, mirroring undergraduate/graduate research supervision.

## RESEARCH PUBLICATIONS

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### Intent Recognition, AI & Autonomy

1. **Md Abu Sayed**, Md Azizul Hakim, Ayesh Meepaganithage, Monica Nicolescu, and Mircea Nicolescu. Feature-aware deep learning for maritime intent recognition. In *IEEE International Conference on Future Machine Learning and Data Science (FMLDS 2025)*, 2025. forthcoming
2. **Md Abu Sayed**, Md. Azizul Hakim, Ayesh Meepaganithage, Tyler J Becker, Monica Nicolescu, and Mircea Nicolescu. Early classification of intentions for maritime domains using deep learning models. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, pages 1171–1178, 2025
3. Ayesh Meepaganithage, **Md Abu Sayed**, Mircea Nicolescu, and Monica Nicolescu. Proactive maritime threat prediction: Vessel intent classification with lstms and transformers using a sliding window approach. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE 2025)*, pages 3261–3267, 2025
4. Mayamin Hamid Raha, **Md Abu Sayed**, Monica Nicolescu, Mircea Nicolescu, and Sushil Louis. Ram: Resource allocation for multi-agent maritime environment. In Giuseppina Gini, Henk Nijmeijer, and Dimitar Filev, editors, *Informatics in Control, Automation and Robotics*, pages 69–95, Cham, 2026. Springer Nature Switzerland
5. Mayamin Raha, **Md Abu Sayed**, Monica Nicolescu, Mircea Nicolescu, and Sushil Louis. Maritime dynamic resource allocation and risk minimization using visual analytics and elitist multi-objective optimization. In *Proceedings of the 20th International Conference on Informatics in Control, Automation and Robotics - Volume 1: ICINCO*, pages 54–63. INSTICC, SciTePress, 2023

### Simulation, Maritime AI & Multi-Agent Systems

6. **Md Abu Sayed**, Korben DiArchangel, Mayamin Hamid Raha, Parvaneh Aliniya, Monica Nicolescu, Mircea Nicolescu, and Sushil Louis. Navysim: A multi-vessel simulation and analysis engine for naval domains. In *2024 IEEE Conference on Games (CoG 2024)*, pages 1–8, 2024
7. **Md Abu Sayed**, Korben DiArchangel, Parvaneh Aliniya, Casey Brenner, Mayamin Hamid Raha, Monica Nicolescu, Mircea Nicolescu, and Sushil Louis. Navysim 2.0: Enhanced multi-vessel simulation and analysis engine for advanced naval research. *IEEE Transactions on Games*, 2025. Submitted
8. **Md Abu Sayed**, Mayamin Hamid Raha, Parvaneh Aliniya, Monica Nicolescu, and Mircea Nicolescu. Threatmap: A framework for enhancing security awareness and decision-making for naval agents. In *Proceedings of the 26th International Conference on Harbor, Maritime and Multimodal Logistic Modeling & Simulation (HMS 2024)*. CAL-TEK, 2024
9. Mayamin Hamid Raha, **Md Abu Sayed**, Sergiu Dascalu, Monica Nicolescu, and Mircea Nicolescu. Keep sailing: An investigation of effective navigation controls and subconscious learning in simulated maritime environment. In *International Conference on Information Technology-New Generations*, pages 313–322. Springer, 2024

### Medical Imaging & Computer Vision

10. **Md Abu Sayed**, Sajib Saha, GM Atiqur Rahaman, Tanmai K Ghosh, and Yogesan Kanagasingam. An innovate approach for retinal blood vessel segmentation using mixture of supervised and unsupervised methods. *IET Image Processing*, 15(1):180–190, 2021
11. **Md Abu Sayed**, Sajib Saha, GM Rahaman, Tanmai K Ghosh, and Yogesan Kanagasingam. A semi-supervised approach to segment retinal blood vessels in color fundus photographs. In *Conference on Artificial Intelligence in Medicine in Europe*, pages 347–351. Springer, 2019
12. Tanmai K Ghosh, Sajib Saha, GM Rahaman, **Md Abu Sayed**, and Yogesan Kanagasingam. Retinal blood vessel segmentation: A semi-supervised approach. In *Iberian Conference on Pattern Recognition and Image Analysis*, pages 98–107. Springer, 2019

## RESEARCH & PROJECTS

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- NavySim: Multi-vessel Simulation Engine** | Unity, Shader, Python 2022 – Present
- Built a scenario-generation pipeline enabling physics-consistent multi-agent maritime interactions.
  - Integrated threat heatmaps and real-time ML-based intent recognition into the simulation.
- Intent Recognition for Maritime Autonomy** | *HMM, LSTM, Transformers* 2022 - Present
- Developed probabilistic and deep-learning intent recognition models (HMMs, LSTMs, Transformers) for early prediction of vessel behavior in complex maritime scenarios.
  - Designed feature-aware models incorporating bearing-rate, CPA/TCPA, and motion derivatives to improve early prediction under sparse and noisy observations.
- Generative Sequence Modeling for Maritime Behaviors** | *CVAE, TimeGAN, ACGAN* 2024 - Present
- Built generative models (CVAE, TimeGAN, ACGAN, LSTM-GAN inspired variants) to forecast future trajectories and reconstruct missing/shadowed maritime sensor data.
  - Designed multi-task generative models that jointly predict future vessel motion and latent intent.
  - Implemented encoder-decoder models for long-horizon forecasting and simulating adversarial naval behaviors.
- Role Shifting in Human-Robot Collaboration** | *Embodied Perspective-Taking* 2025 - Present
- Applied ASL-inspired torso-based role shifting to improve perspective and referential clarity in robot communication.
  - Implemented and compared gesture-only vs. torso-rotation communication behaviors on a Unitree G1 courier task.
  - Designed evaluation metrics to assess clarity, naturalness, and perceived collaborative intelligence in HRI.
- ThreatMap: Maritime Situational Awareness** | Python, Unity, Shader 2021 – 2024
- Developed visual and mathematical models for estimating dynamic threat levels in multi-vessel environments.
  - Designed a heatmap framework integrating proximity and kinematic cues for real-time visualization.
- Breast Cancer Detection via Multimodal Fusion** | PyTorch 2022 – 2024
- Built graph-based neural models to reason over bilateral and ipsilateral mammogram views for malignancy detection.
  - Implemented tailored augmentation strategies to improve sensitivity to temporal and structural breast tissue changes.
- Retinal Vessel Analysis & Local Haar Pattern (LHP)** | *Python, MATLAB* 2018 – 2020
- Developed segmentation pipelines and the Local Haar Pattern descriptor for retinal fundus images, capturing domain-specific vascular and lesion structures.
  - Engineered vessel enhancement, feature extraction, and morphological post-processing and used SVM/Random Forest models for vessel vs. non-vessel classification.

## TECHNICAL SKILLS

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**Languages:** Python, C/C++

**ML/AI/CV Frameworks:** TensorFlow, Keras, PyTorch, OpenCV

**Simulation/Game:** Unity, C#

**Databases:** PostgreSQL/pgvector, MySQL

**Web & Tools:** FastAPI, React, Git, Replit, Figma

## EXTRACURRICULAR ACTIVITIES

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- Graduate Student Association** | Council Member 2024 – 2025
- Chair, Awards Committee; contributed to review of university-wide nominations.
  - Supported graduate student programming through Fundraising & Community Relations and Service Committees.
- International Students Club** | Vice President 2023 – 2024
- Organized Night of All Nations (600 participants) and monthly community-building events.
  - Collaborated with campus organizations to enhance inclusivity initiatives and provide resources for international and multilingual student communities.
- Google Developer Group Campus** | Co-Lead 2024 – 2025
- Co-organized DevFest Reno, introducing students to AI/ML development tools.
- Bangladeshi Student Association** | Treasurer 2022 – 2023
- Managed financial operations and cultural programming for the organization.